Experiments in Fluids (2018) 59:63
https://doi.org/10.1007/s00348-018-2517-0

RESEARCH ARTICLE

Experimental study on rotating instability mode characteristics of axial compressor tip flow

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Received: 26 October 2017 / Revised: 24 February 2018 / Accepted: 27 February 2018
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Abstract

This paper investigates the rotating instabilities that occurred on the single-stage axial compressor designed for aerodynamic performance validation, which was tested with two sets of circumferential measuring points in combination. Circumferential mode characteristics of compressors are usually too high to be captured experimentally, and aliasing of the circumferential mode order occurs when not enough sensors are used. A calibration and prediction method to capture the higher circumferential mode of unsteady flow in a compressor was proposed. Unsteady pressure fluctuations near the tip region in an axial compressor were studied, and high circumferential mode characteristics were captured on both the blade passing frequency (BPF) and the rotational instability frequency (RIF) under different flow rate conditions based on this novel method. The characteristic RI spectrum with a broadband hump was present in a large range of flow conditions. Both the frequency range and the dominant circumferential mode order decreased as the flow rate decreased. Based on the calibrated mode characteristics, a rotating aerodynamic source model is used to explain the side-by-side peak of RIF spectrum and rotating characteristics of RI. The calibration and prediction method of the high circumferential mode is beneficial for the research of unsteady flow in an axial compressor.

List of symbols

B	Number of rotor blades
V	Number of inlet guide vanes
f	Frequency
p	Pressure
t	Time
m	Circumferential mode order
m_x	Actual mode order
m'	Transition mode order
N	Number of evenly spaced measuring points in a ring
φ	Flow coefficient
ω	Angular velocity
ω_m	Angular velocity of the rotating mode
θ_s	Angular position
ψ	Total pressure coefficient
$\Delta\phi$	Circumferential phase difference
η	Efficiency
Ω_R	Angular velocity of rotor
Ω_m	Angular velocity of mode *m at BPF*
LCM	Least common multiple
Ω_{Q}^{F}	Angular velocity of source Q in the fixed reference frame
ω_{m}^{F}	Angular frequency of mode wave component *m*
ω^{Q}	Angular frequency of the aerodynamic source Q
n_Q^F	Source rotating speed
f_{m}^{F}	Frequency of mode wave component m
f^{Q}	Frequency of the aerodynamic source Q

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1 Introduction

Rotating instability (RI) is a flow phenomenon that occurs near the tip region of fan and compressor at a relatively small flow rate and a large tip clearance design. It represents a local aerodynamic instability and is potentially connected to unsteady tip leakage flow. RI can be related to a characteristic spectrum with a wide-band hump below the blade passing frequency (BPF) in pressure, velocity or noise measurements.

Mathioudakis and Breugelmans (1985) first described this phenomenon in a single-stage axial compressor as "small rotating stall". In particular cases, RI could be associated

Published online: 07 March 2018

with blade vibrations (Baumgartner et al. 1995; Holzinger et al. 2016) and a significant increase in tip clearance noise (Kameier and Neise 1997). A consensus has not been reached on the generation mechanism of RI. Mailach et al. (2001) and März et al. (2002) recognized a periodic interaction of the tip clearance flow of one blade and the flow at the adjacent blade to be responsible for RI, whereas other research focused on the influence of backflow (Vo 2010) and shear layer instabilities (Pardowitz et al. 2012, 2015). Considering RI and tip clearance noise, Neuhaus and Neise (2002, 2003) introduced air injections into the tip clearance gap to improve aerodynamic and acoustic performance.

The rotating instability phenomenon includes a circumferential propagating characteristic with a superposition of circumferential pressure mode waves, which are similar to azimuthal sound modes in ducts. Holste and Neise (1995) measured the pressure fluctuations in the exit plane of the CRISP (counter rotating integrated shrouded propfan) model in 1995, resolving the duct modes of tone components with results on 120 evenly spaced angular positions to determine the dominant noise sources. Liu et al. (1996) studied the azimuthal structure of RI in axial turbomachines using a similar numerical mode analysis method with 72 evenly spaced circumferential measurement positions and 7 microphones. Their research showed that there was only one dominant mode at each peak frequency of the rotating instability components, and subsequent modes were continuously numbered. Recently, Pardowitz et al. (2012, 2015) investigated the RI phenomenon experimentally at an annular stator cascade and an axial fan stage with a maximum number of 72 (near field) and 16 (far field) evenly spaced sensors, respectively, which showed a refined mode analysis. In addition, a time-resolved analysis was used and provided new insight into the unsteady properties of the RI.

According to previous research, measurement and circumferential mode analysis of compressor tip flow are important to study complex flow characteristics. To analyse flow phenomena with relatively high circumferential mode order, expensive investment in enough numbers of pressure/microphone sensors and the consequent sampling system is usually required. In this paper, a calibration and prediction method of higher circumferential mode order is proposed, and two sets of variable, evenly spaced sensors are used simultaneously to overcome mode aliasing and capture the flow characteristics of relatively high circumferential mode order using much fewer sensors. The pressure fluctuations caused by rotor–stator interaction and the RI phenomenon were measured and analysed with this novel method in the single-stage axial compressor. A comparison of results at different flow conditions showed the evolution of RI. Moreover, a rotating aerodynamic source model of RI is presented using the calibrated mode characteristics, which explains the side-by-side peak of RIF spectrum and rotating characteristics of RI.

2 Experimental setup

The experimental investigation is performed on the subsonic single-stage axial compressor rig of Shanghai Jiao Tong University. The impeller diameter is 598 mm, with a hub-to-tip ratio of 0.7. The rotor is equipped with 21 NACA 65 blade profiles with a chord length and stagger angle at mid-span of 60 mm and 35°, respectively. The inlet guide vanes consist of 13 blades, and the axial distance between the rotor and IGV at mid-span is approximately 110% of the rotor blade chord. The design flow rate and rotating speed are 4.9 kg/s and 3000 rpm, respectively. The rotor tip clearance is set as 2.2% of the blade height because the occurrence of the RI phenomenon under specified operation conditions was captured in previous studies.

The aerodynamic experiments of the compressor are carried out according to ISO5801 (1997E), the test methods of aerodynamic performance for fans. A side view of the test section with its main components is shown in Fig. 1. At measurement plane I, a pneumatic probe is used to acquire the inlet total temperature and total pressure together with the inlet wall static pressure achieved by static pressure holes imposed on the casing wall, by which the flow rate of the compressor is calculated. A five-hole probe is traversed

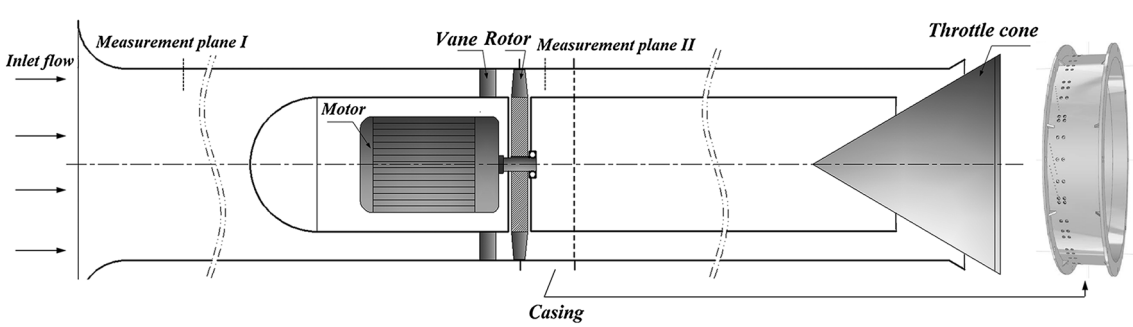


Fig. 1 Subsonic single-stage compressor test rig

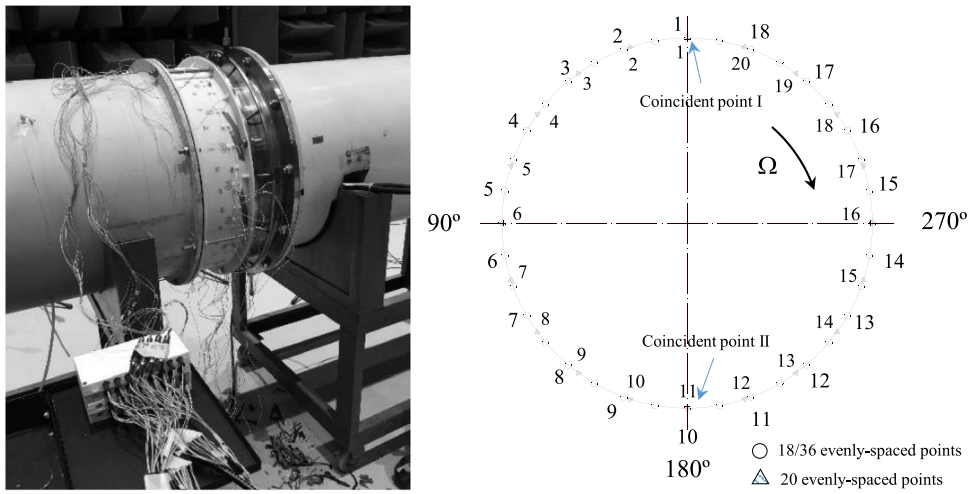


Fig. 2 Experimental configuration of pressure sensor

in the radial direction at measurement plane II to acquire the spanwise distributions of total pressure and flow angle downstream the rotor. A throttle cone in the downstream duct is adopted to adjust the aerodynamic operating condition. Detailed introduction is shown in Wang et al. (2016a, b, c).

Thirty-six Kulite XCQ-093 dynamic pressure transducers are circumferentially flush-mounted on the casing wall to measure the unsteady pressure fluctuation of the rotor tip flow. The axial position of the sensor ring is 1/4 chord length downstream from the rotor blade trailing edge. Three sets of NI PXIe-4499 Modules with 24 bits and a maximum sample rate of 204.8 kS/s are adopted for signal collection.

A total of 36 or 20 pressure sensors are situated in one ring with equidistant circumferential spacing, which can achieve a synchronous measurement of 36 points or 18 and 20 points (both requiring 36 sensors), as shown in Fig. 2. The calibration method for identifying the high circumferential mode order of rotor tip flow is described in Sect. 3.

The aerodynamic performance map of the single-stage axial compressor at design rotating speed $n = 3000$ rpm is shown in Fig. 3, which is averaged result from three-times aerodynamic experiment. The design flow coefficient is $\varphi = 0.151$. In the present experiment, blade tip pressure fluctuations are measured under 16 different flow coefficients varying from $\varphi = 0.172$ to $\varphi = 0.084$, as listed in Table 1.

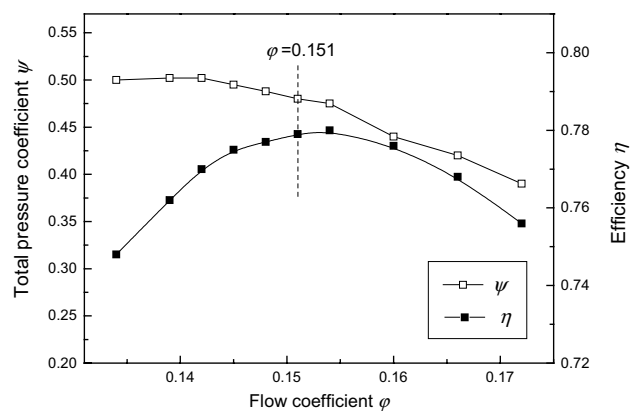


Fig. 3 Compressor performance map at 3000 rpm

Table 1 Flow rate conditions

No.	Flow coefficient φ	No.	Flow coefficient φ
1	0.172	9	0.139
2	0.166	10	0.134
3	0.160	11	0.128
4	0.154	12	0.121
5	0.151	13	0.113
6	0.148	14	0.10
7	0.145	15	0.091
8	0.142	16	0.084

3 Circumferential modes calibration method

3.1 Mode decomposition

The mode decomposition method based on spatial Fourier transformation is generally adopted to transform the signals from the spatial domain to the circumferential wave number (also defined as the circumferential mode order) domain. The diagram of the mode decomposition process is depicted in Fig. 4, where the signal consists of pressure waves of two mode order with a different rotating direction. For different analysis purposes, pressure signals in the time domain and frequency domain can be decomposed into several circumferential modes (Vo 2010; Pardowitz et al. 2012).

Table 2 Calibrating the mode order at BPF

Calibrated mode order	Measuring points	Initial results	$v=1$	$v=2$	$v=3$	$v=-1$	$v=-2$	$v=-3$
21	18	3	**21**	39	57	− 15	− 33	− 51
	20	1	**21**	41	61	− 19	− 39	− 59
	36	− 15	**21**	57	93	− 51	− 87	− 123
− 5	18	− **5**	13	31	49	− 23	− 41	− 59
	20	− **5**	15	35	55	− 25	− 45	− 65
	36	− **5**	31	67	103	− 41	− 77	− 113

Bold values indicate the recurring mode orders of transition mode m' for $-50 < m' < 50$

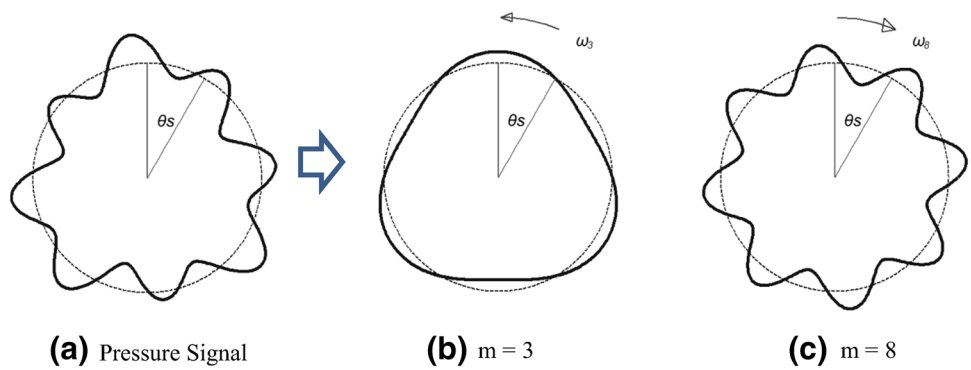


Fig. 4 Mode decomposition process

The circumferential distribution of the unsteady pressure signals next to the rotor trailing edge is represented in both the time domain and frequency domain at angular position θ_s,

$$p(\theta_s, t) = \int_{-\infty}^{\infty} p(\theta_s, f) e^{2\pi i f t} \mathrm{d}f \tag{1}$$

$$p_s(\theta_s, f) = \sum_{m=-\infty}^{+\infty} P(m, f) e^{im\theta_s} \tag{2}$$

Thus:

$$p(\theta_s, t) = \int_{-\infty}^{+\infty} \sum_{m=-\infty}^{+\infty} P(m, f) e^{im\theta_s} e^{2\pi i f t} df = \sum_{m=-\infty}^{+\infty} P(m, t) e^{im\theta_s} \tag{3}$$

where $P(m, t)$ represents the pressure amplitude of mode m at time t.

The discrete forms of mode decomposition may also be written as,

$$P(m, f) = \frac{1}{N} \sum_{s=0}^{N-1} p(\theta_s, f) \cdot e^{-im\theta_s} \tag{4}$$

$$P(m, t) = \frac{1}{N} \sum_{s=0}^{N-1} p(\theta_s, t) \cdot e^{-im\theta_s} \tag{5}$$

where N is the number of equidistant measuring points on each circumferential plane and m is the circumferential mode order. A positive mode order corresponds to a pressure wave travelling in the same rotational direction of the rotor and vice versa.

In particular, mode decomposition in time domain $P(m, t)$ involves the time-resolved characteristics of different modes of the pressure wave, with the amplitude representing its strength over time and the phase representing its spatial phase relative to angular position $\theta_s = 0°$, which may be written as,

$$P(m, t) = |P(m, t)| e^{i \cdot [\mathrm{m}\omega_m t + \phi_m(t_0)]} \tag{6}$$

where ω_m is the angular velocity of mode m. $P(m, t)$ may be used to reveal propagation characteristics of the pressure fluctuation in the time domain.

3.2 Calibration method of higher mode order

For the conventional mode decomposition method, it is uncertain whether the number of pressure sensors is sufficient to identify the circumferential mode order, and as many sensors as possible are used in synchronous measurement. However, the number of pressure sensors is restricted because of the finite circumferential space of the casing wall. In this section, a novel mode decomposition method is proposed to identify higher mode orders with fewer circumferential sensors.

The law of aliasing in the circumferential mode order domain is introduced to reveal the defect of the present mode decomposition method.

Consider pressure signals $p(\theta)$ at different angular positions θ for time t_i. The signal can be simplified as an ideal sampling process,

$$p_s(\theta) = p(\theta) \cdot s(\theta) = p(\theta) \cdot \sum_{n=-\infty}^{\infty} \delta(\theta - n\Delta\phi) = \sum_{n=-\infty}^{\infty} p(n\Delta\phi) \cdot \delta(\theta - n\Delta\phi) \tag{7}$$

where $s(\theta)$ is a pulse function related to sampling and $\Delta\phi$is the sampling interval in the circumference, $\Delta\phi = 2\pi/N$.

According to the convolution theorem of spatial Fourier transformation,

$$P_s(jm) = \frac{1}{\Delta\phi} \sum_{k=-\infty}^{\infty} P\left(jm - kj\frac{2\pi}{\Delta\phi}\right) \tag{8}$$

Equation (8) shows that a translation and superposition of a continuous signal in a circumferential wave number domain formed the sampling result. In practical signal processing, the range of a circumferential wave number domain is $-N/2 < m \leqslant N/2$ according to the Nyquist sampling theory. Figure 5 shows sketches of discrete wavenumber signal sampling; no aliasing is depicted in Fig. 5a, and the dotted line indicated by the arrow represents an aliasing phenomenon in Fig. 5b.

For a translation amplitude of $2\pi/\Delta\phi$, the signal with actual mode order m_x varies in the mode decomposition result as,

$$m = m_x + v \cdot \frac{2\pi}{\Delta\phi} = m_x + v \cdot N, v = 0, \pm1, \pm2 \cdots \tag{9}$$

Holste (Holste and Neise 1997) described a similar formation. Equation (9) explains why flow phenomena with higher mode order are difficult to identify in the mode decomposition result. In the present experiment, two sets of equidistant circumferential spacing sensor numbers labelled N_1 and N_2 are adopted, and the actual mode is written as,

$$m_x = m_1 - v_1 \cdot N_1 = m_2 - v_2 \cdot N_2, \quad v_1, v_2 = 0, \pm1, \pm2 \cdots \tag{10}$$

Here, m_1, m_2 are the mode decomposition results corresponding to sensor numbers N_1 and N_2, respectively, according to the Nyquist sampling theory.

Under the special case $m_1 = m_2$, Eq. (10) simplifies to

$$m_x = m_1 + i \cdot S = m_2 + i \cdot S, \quad i = 0, \pm1, \pm2, \cdots \tag{11}$$

where $S = LCM(N_1, N_2)$ represents the least common multiple of N_1 and N_2. The interval between the actual mode order and its measurement value is enhanced from N_1 and N_2 to S by two sets of measuring positions.

In general, $m_1 \neq m_2$, and a transition mode with minimum absolute value m' is defined as:

$$m' = m_1 - v_1' N_1 = m_2 - v_2' N_2 \tag{12}$$

where (v_1', v_2') is the special integer satisfying Eq. (12).

By subtracting Eq. (12) from Eq. (10), the actual mode m_x can be obtained,

$$m_x - m' = (v_1 - v'_1)N_1 = (v_2 - v'_2)N_2 = i' \cdot S, \quad i' = 0, \pm1, \pm2, \cdots \tag{13}$$

Equation (13) shows that the interval between the actual mode order m_x and the transition mode order m' is an integer multiple of S, as with the enhancement of special case $m_1 = m_2$. That is, with the numbers of sensors being $N_1 + N_2$, the maximum resolved mode order is significantly enhanced from $(N_1 + N_2)/2$ to $LCM(N_1, N_2)/2$.

When considering experimental data, a set of possible actual mode orders can be listed by Eq. (11) for the case of $m_1 = m_2$. If $m_1 \neq m_2$, a transition mode order m' is determined first by Eq. (12), and then a similar set of possible actual mode orders can be listed by Eq. (13). The last step is to pick an element in this set as the actual mode order based

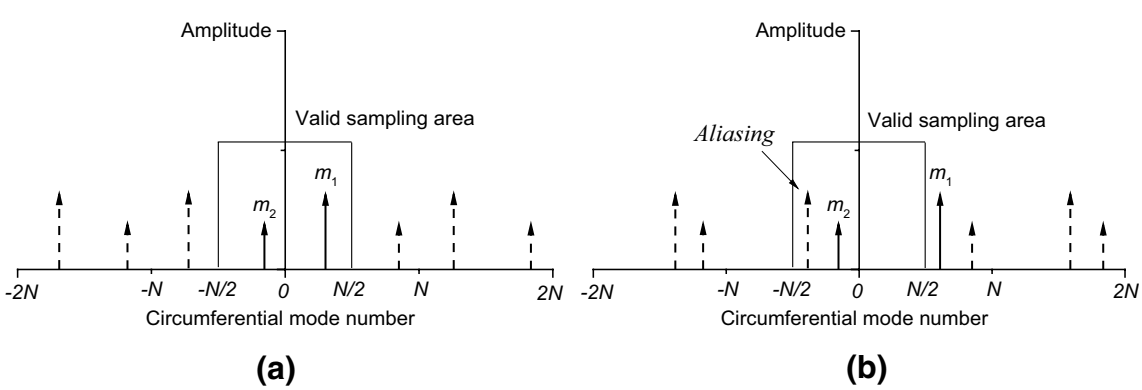


Fig. 5 Discrete wavenumber signal sampling

on prior knowledge, which is to make sure the interested mode order being smaller than the maximum resolved mode order of the calibration method. Compared to simply using Eq. (9), a more effective calibration of high mode order is thus achieved. Higher circumferential mode order is identified based on the fewer measuring points of N_1 and N_2.

4 Results and discussion

The spectral and mode characteristics of RI were investigated based on the unsteady pressure signals of a circumferential distributed sensor ring. The circumferential mode calibration method discussed in Sect. 3 was applied to resolve the RI circumferential characteristics in the frequency and time domain. Also the structure of RI in flow field was analysed by using a rotating aerodynamic source model, which explains the characteristic RIF spectrum.

4.1 Pressure fluctuation characteristics

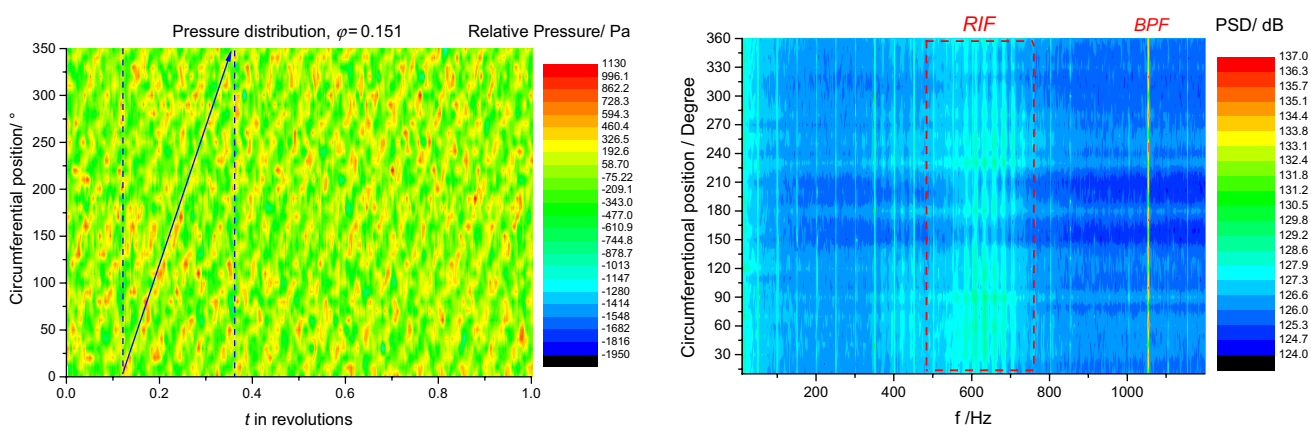
Figure 6 shows the spatial distribution of circumferential pressure at a design flow rate of $\varphi = 0.151$ in the time domain and frequency domain. Figure 6a depicts an apparent circumferential propagation progress of pressure fluctuation with almost constant propagating speed. The pressure fluctuation has 21 cycles in one revolution, which can be matched to the blade passing frequency (BPF). Moreover, the pressure fluctuation at the same time shows 5 strips in the circumferential position axis, and its angular velocity is approximately $-4.25\ \Omega_R$ in fixed reference frame. This phenomenon can be attributed to unsteady interaction between the IGVs and the impeller. Figure 6b shows a similar frequency domain pressure distribution at different circumferential positions, which is also an underlying hypothesis of the traditional mode decomposition method. Notice that the range of rotating instability frequency (RIF) indicated by the dashed line is approximately 0.5–0.7 BPF.

Pressure PSD spectra in all 16 flow conditions are shown in Fig. 7. At an RI frequency range below the 1st BPF, a wide-band amplitude hump rises as the flow rate decreases, and gradually, a characteristic side-by-side peak phenomenon occurs in the PSD spectra; at the same time, RIF moves to a relatively low frequency. In small flow conditions ($\varphi = 0.128$–0.084), the wide-band hump characteristic changes, with an obvious single peak and fixed frequency range near 700 Hz.

Characteristic PSD spectra of RIF in two different flow conditions are shown in Fig. 8. Each spectrum involves the results of two different axial positions. A difference in shape between the previous wide-band hump and the subsequent single peak is observed. In addition, RI decreases in amplitude after downstream propagation only at relatively large flow rates. There is no other notable difference between the results for the two axial positions.

4.2 Mode decomposition and calibration

The mode analysis methods described in Sect. 3 are applied to the measured data. Three main forms of periodic pressure fluctuation can be related to particular circumferential mode characteristics of the pressure spectrum, and mode decomposition result without calibration is shown in Fig. 9. Rotor frequency (RF) here means a pressure fluctuation caused by imbalance of the rotor, with its mode order directly proportional to frequency and starting with $m = 1, 2 \ldots$; RIF usually consists of a few continuous numbered mode waves; BPF is related to the rotor blades and their interaction with the inlet guide vane, in this case $m = -5, -15$ are found to be the two main mode number at this frequency, as shown in Fig. 9b.



(a) Spatial-time domain **(b)** Spatial-frequency domain

Fig. 6 Distribution of spatial pressure fluctuation

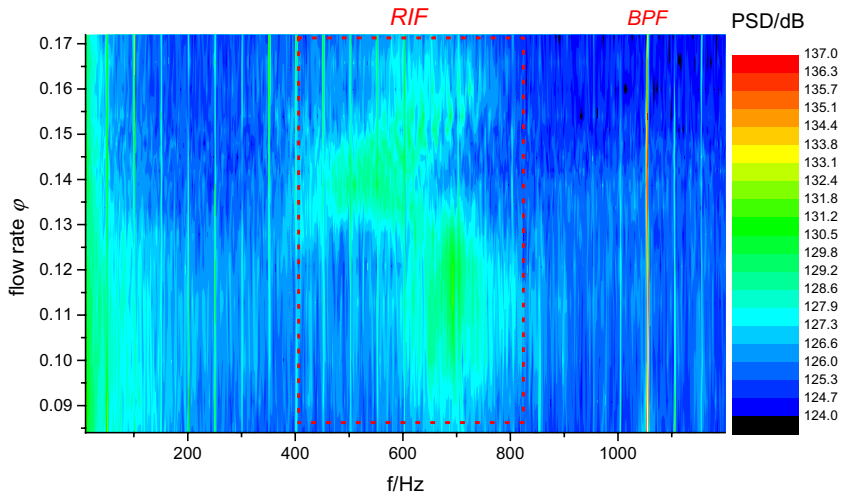


Fig. 7 PSD in different flow conditions

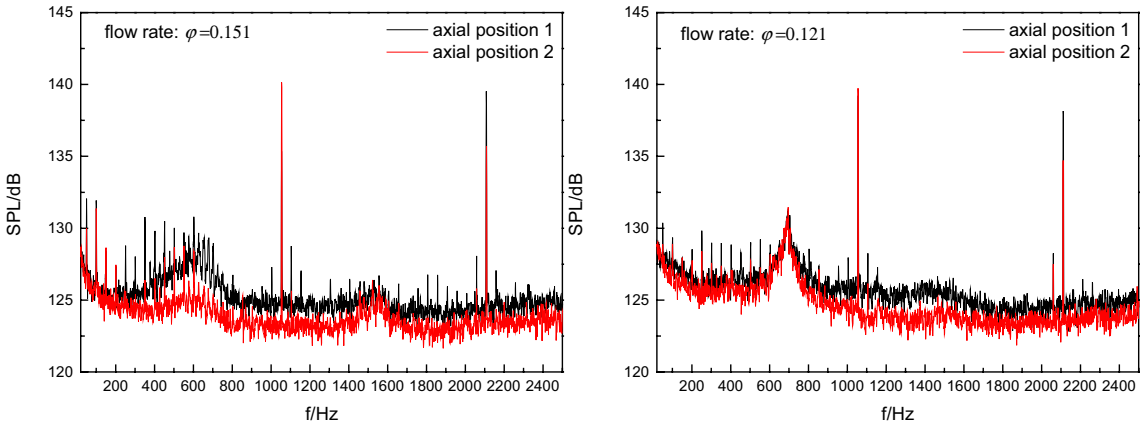
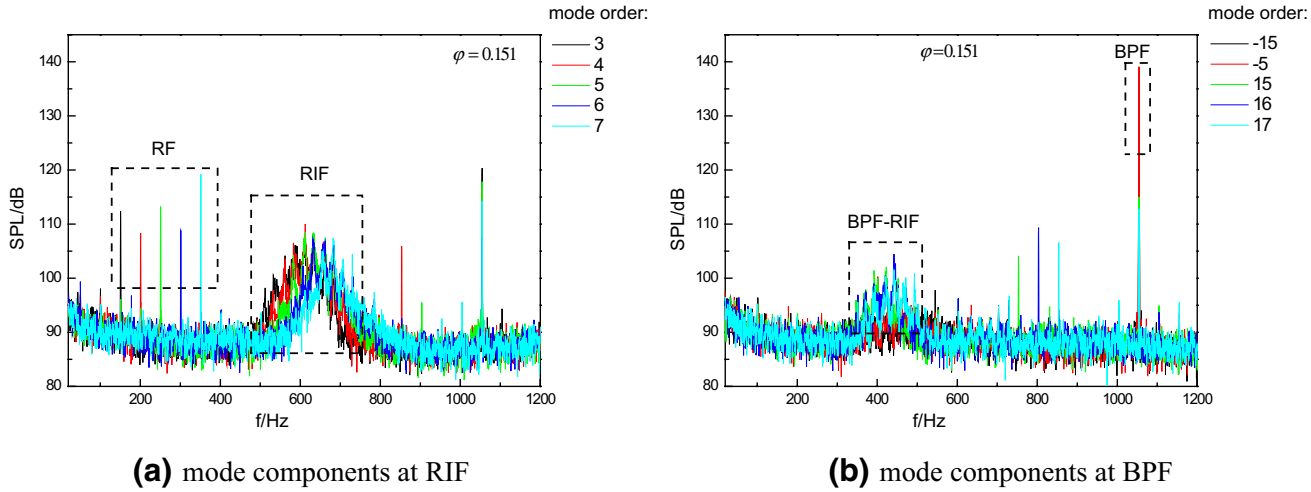


Fig. 8 Characteristic PSD spectra at different axial position



(a) mode components at RIF **(b)** mode components at BPF

Fig. 9 Circumferential mode components in PSD

Interaction between the rotor blades and RI forms another wide-band hump.

An analysis of the circumferential mode components at BPF is applied here to validate the mode calibration method. Two dominant circumferential mode orders at BPF can be obtained with results of 18, 20 and 36 measuring positions, as shown in Fig. 10. An amplitude difference can be seen for the corresponding mode orders. This can be explained as an unavoidable error in amplitude due to the aliasing phenomenon with a decrease in measuring points.

Equation (9) is applied, and an enumeration of the possible mode orders at BPF is shown in Table 2. Two calibrated mode orders, "21" and "− 5", are determined by this method. For each case, there is only one transition mode order m' for $-50 < m' < 50$, which is $m' = -5$ and $m' = 21$. The relation between an original mode order and its transition mode order is $m_x = m' - i' \cdot S$. As a dominant circumferential mode order of more than 180 (the least common multiple of 18 and 20) is unlikely at BPF, "21" and "− 5" can be considered the actual mode. Also it can be seen that the calibration method will be invalid if only using results with 18 and 36 measuring points, for they don't have a big enough least common multiple number.

According to Tyler and Sofrin (1962), the circumferential mode order and corresponding angular velocity in fixed reference frame can be expressed as:

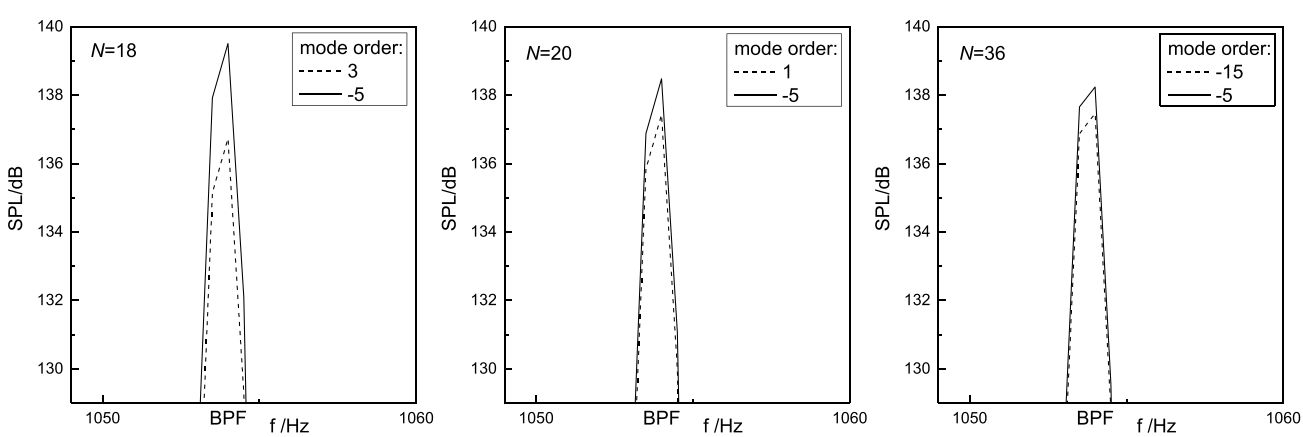
$$m = nB + kV, \Omega_m = \frac{nB}{nB + kV}\Omega_R \tag{14}$$

For the single-stage compressor, rotor blade number is $B = 21$, and the number of inlet guide vanes is$V = 13$. The dominant mode order caused by interaction between rotor blades and vanes is $m = -5$ ($n = 1,\ k = -2$), and its angular velocity is $\Omega_m = -4.2\Omega_R$, which is similar to the angular velocity of the pressure fluctuation observed in Fig. 6a. Another mode order $m = 21$ ($n = 1,\ k = 0$) represents the pressure fluctuation of the rotor blades.

As validated by mode analysis of rotor–stator interaction phenomenon, the calibrated mode analysis method is applied to RIF results. Different mode decomposition results are achieved for relatively large and small flow conditions.

The mode order composition of RIF under eight flow conditions are shown in Fig. 11. The wide-band hump of RIF has a relatively small amplitude under large flow rate conditions in Fig. 11a ($\varphi = 0.172$–0.166). In the next few flow rate conditions ($\varphi = 0.160$–0.148), RIF develops significantly in amplitude, and the spectrum of each mode order is shown with multiple peaks, which makes it difficult to identify a dominant mode order for each peak of the origin spectrum, as shown in Fig. 11b, c. With a further decrease in the flow rate ($\varphi = 0.145$–0.134), RIF shifts to a relatively low frequency, and a clear continuous mode order distribution emerges at the peak of each mode order, as shown in Fig. 11d–f. Also the frequency range shift between flow conditions is more obvious here.

In small flow rate conditions of in Fig. 11g, h ($\varphi = 0.128$–0.121), the wide-band hump of RIF degenerates. A total frequency shift of nearly 300 Hz is observed for this stage of RI development. Although mode order composition of single-peak RI in small flow conditions is different, only mode orders 1, 2 and 3 are found in the decomposition results, which are not the interaction results between BPF and RI. The shapes of these low mode order peaks are almost the same as the flow rates changed from 0.128 to 0.121. The laser displacement sensor is used to measure the natural frequency of the rotor, which is the same as the low mode order peak found at 0.128 and 0.121 flow rate conditions. The natural frequency is found at these two low flow



(a) 18 measuring positions **(b)** 20 measuring positions **(c)** 36 measuring positions

Fig. 10 Circumferential mode components at BPF

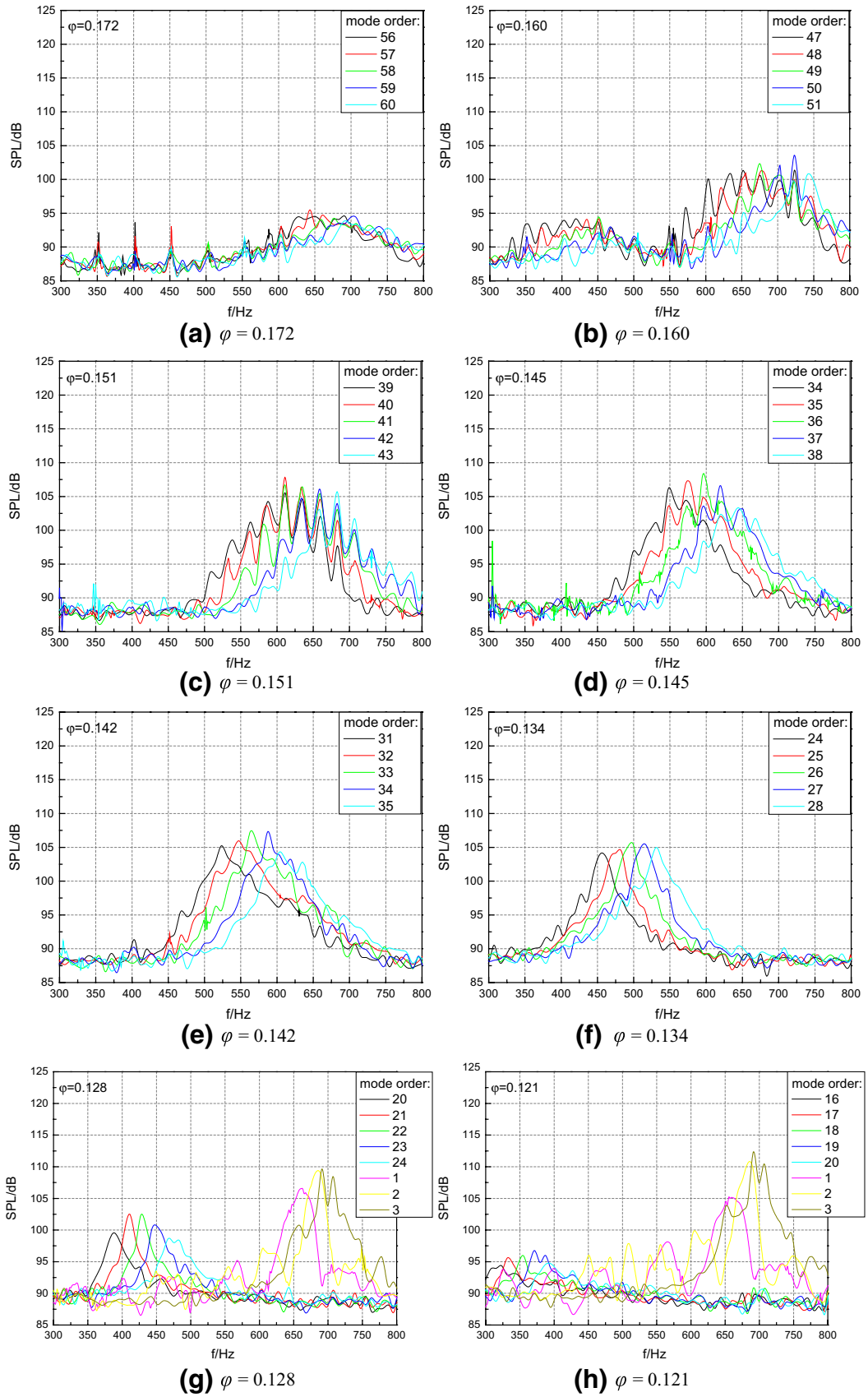


Fig. 11 Circumferential mode decomposition results at RIF (different flow rate)

rates and not at high flow rates such as at 0.172 and 0.166, indicating that the high mode RI flow may not have a direct influence on the resonance of the rotor.

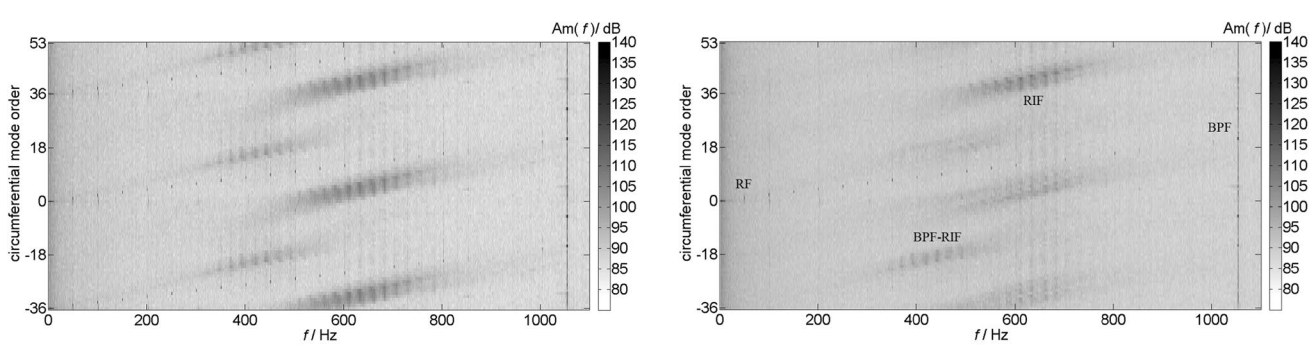
The original and calibration mode decomposition results are compared in Fig. 12. In Fig. 12a, the original mode decomposition results were stacked according to Eq. (9) using results of 36 measuring points. The stacked results with two sets of measuring points (18 and 20) are compared at each mode and each frequency in the program, and only common peak values are retained in the spectrum. What needs to be mentioned is that also results with 36 measuring points are used together in this process for it actually shows a better calibration result. The calibration mode decomposition results are shown in Fig. 12b. Both harmonic modes of rotating frequency (RF) and two dominant modes at BPF can be observed without the aliasing mode orders once appeared in Fig. 12a. Component of RI is also recognized to a certain extent, decreasing in amplitude only in aliasing mode order regions (the colour becomes lighter). The calibration mode decomposition result approximated by the results from more than 100 sensors is thus achieved, which is a large improvement under the existing experimental conditions. But it is also shown that the calibration process performs better in mode components of single frequency (RF& BPF) than in wide-band hump regions (RIF). This is because the calibration method is based on comparing and labelling of the spectrum for each frequency, and it cannot extract different mode components again as the original mode decomposition process did.

The calibrated mode order compositions under different flow rate conditions are shown in Fig. 13. The harmonic modes of RF and two dominant modes of BPF are observed to remain constant among different flow conditions. The RI mode component indicates a frequency shift among the continuous mode components. In addition, the mode component of the interaction between the rotor blades and RI, which is a BPF-RI pattern, can be observed as another oblique region with smaller amplitude. In Fig. 13a, the cloud chart is quite clear which shows RI hasn't fully developed. As the flow rate decreases, the RI-related oblique region moves to relatively low frequency with its strength enhanced at first and then weakened. Also a better calibration results is achieved in Fig. 13d, e with the only oblique region for RI and BPF-RI pattern. In the results for relatively small flow rate conditions, Fig. 13e, f, the disturbance component of the low mode order can be seen, and corresponding to Fig. 11, the mode component of the BPF-RIF pattern is simultaneously close to this disturbance.

In Pardowitz et al. (2012), the mechanism of clearance noise (CN) is generally observed while varying the rotor speed, the operating condition, and the rotor configuration. However, the resulting far-field amplitudes of CN depend on the combination of the RI mode orders together with BPF-RI modes and the cut-on regions in the duct. In the experiment, the RI mode order is reduced and the BPF-RI mode order approaches the zero mode with a decrease in flow rates. At some flow rates, the clearance noise corresponding to the low BPF-RI mode may be captured.

4.3 Rotating characteristics of RI

The RI modes observed in the different flow rate conditions are shown in Fig. 14, each flow rate condition selecting five modes with largest amplitude. Different from Figs. 11 and 13, all the flow rate conditions that can be identified with the existence of RI has been used here. The RI mode orders of RI decrease with flow rate almost linearly from 50 to less than 20; the frequency range of RIF shifts to the low-frequency region. It is found that in relative large flow rate conditions, the frequency of a certain mode order becomes higher when the flow rate decreases. While in relative small flow rate conditions, the frequency of each mode order is stable.



(a) Stack of origin results **(b)** Calibrated results

Fig. 12 Mode decomposition result in cloud chart ($\varphi = 0.151$)

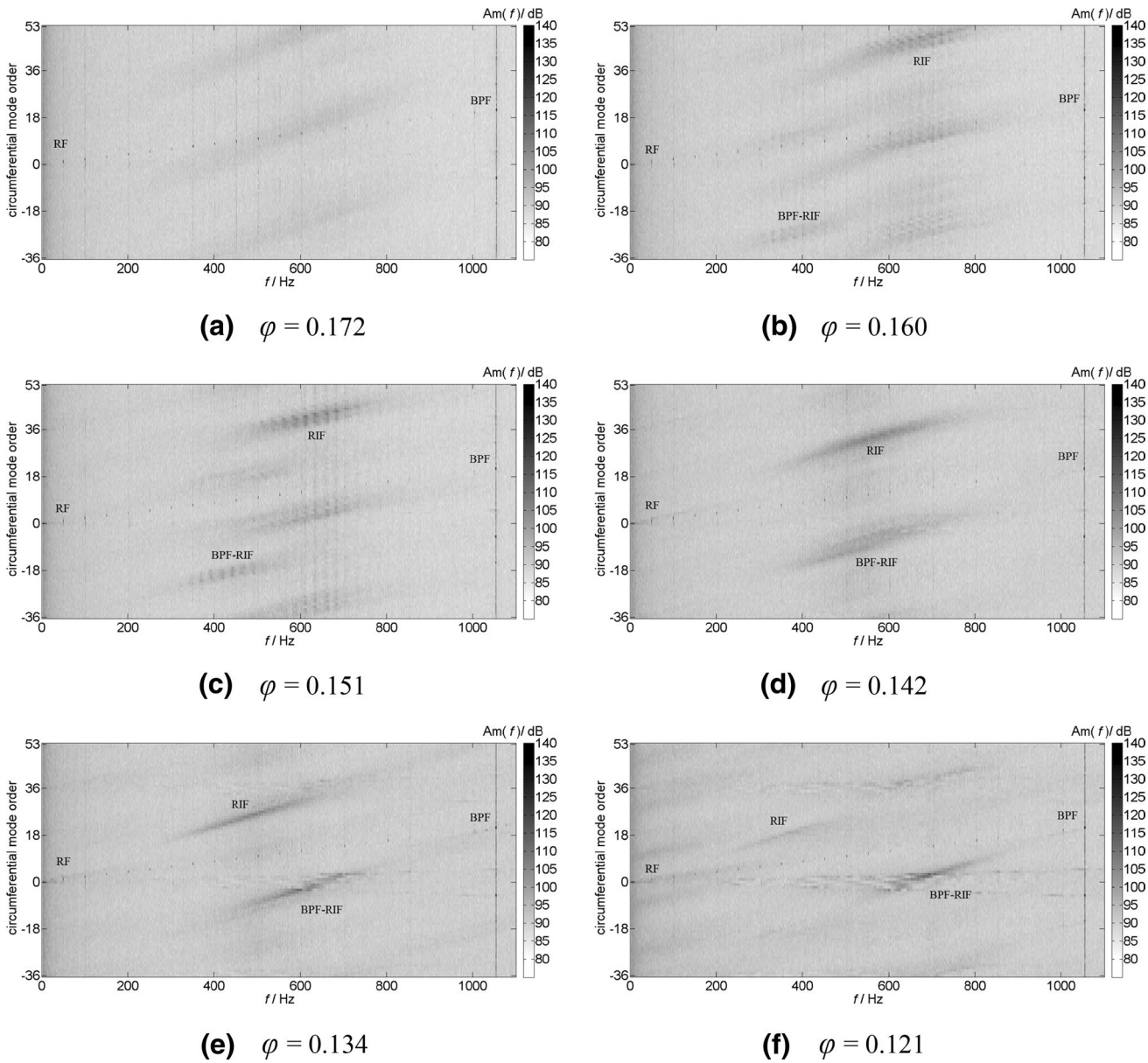


Fig. 13 Calibrated mode decomposition results in cloud chart

The angular velocity of RI modes in fixed reference frame is also needed in order to study the rotating structure of RI phenomenon. In order to better understanding the RI rotating structure, the theory of rotating aerodynamic source mentioned in Ref (Kameier and Neise 1997) is considered in the following analysis. Assuming RI generated as an aerodynamic source Q located in tip flow region, it has the angular frequency of ω^{Q} and rotates at an angular velocity of Ω_Q^F in the fixed reference frame with the superscript F. Transforming angular frequency ω^{Q} into frequency f^{Q} and the relationship between the measurable frequencies of RI mode components f_{m}^Fcan be derived:

$$f_{\mathrm{m}}^F = f^{\mathrm{Q}} + m \cdot n_Q^F \tag{15}$$

where n_Q^F represents the source rotating speed (or rotating frequency) in unit of revolutions per second (or Hz).

Equation (15) shows a linear relationship between f_{m}^F and m, which is coincide with the results in Fig. 14. Use the least square fitting method on the data shown in Fig. 14 to achieve f^{Q} and n_Q^F corresponding to intercept and slope, respectively. The fitting results for different flow rate conditions are shown in Fig. (15). It needs to be noted that all the fitting results of f^{Q} are becoming their opposite number now in Fig. 15a.

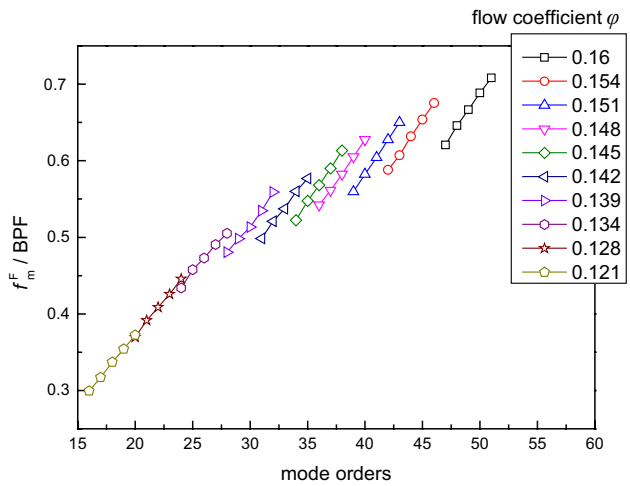


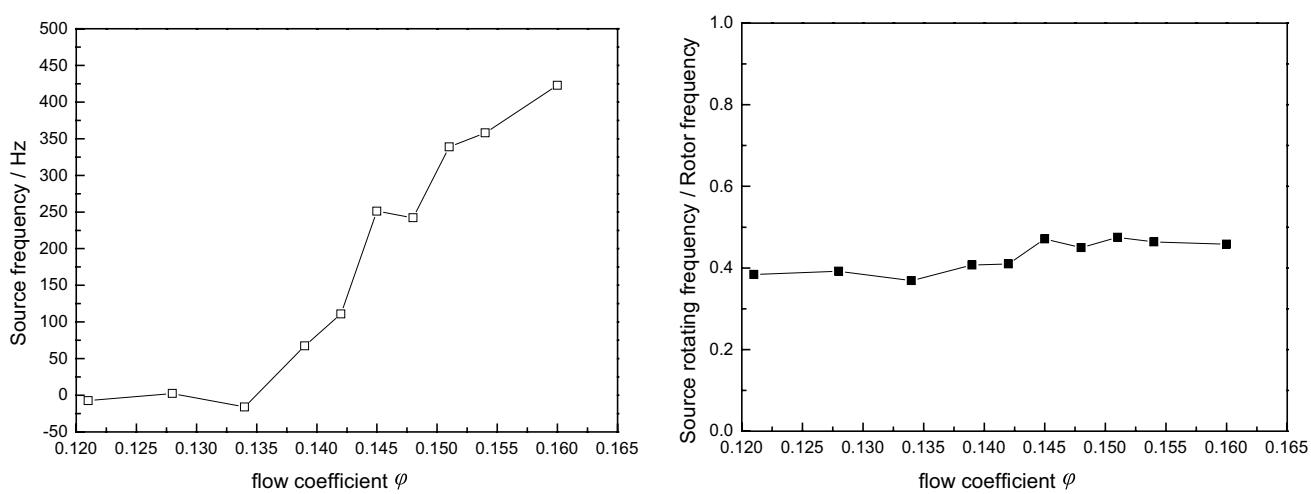
Fig. 14 RI modes in different flow rate conditions

The source frequency reduces when the flow rate decrease while the source rotating speed changes slightly, which is about 40–45% rotor speed. The source frequency is almost zero when the flow rate is fairly small ($\varphi = 0.120$–0.134), which means that the RI aerodynamic source doesn't fluctuating anymore. Also, RIF spectrum characteristics of side-by-side peaks can be considered as a combination of one main peak (with the largest amplitude at RIF) from a certain mode wave that does exist in the flow field and its tunes caused by the rotating of RI source. This can be a reasonable explanation for the side-by-side RIF being continuously numbered in circumferential mode orders, equidistance distributed and reduced in amplitude from the central mode order to both sides. The RI phenomenon is thus considered as an aerodynamic source with a certain source frequency and rotating speed of a single circumferential mode wave. Further validation work of this model is needed to study the existence and flow pattern of the fluctuating aerodynamic source. Results in numerical work might be useful for it to monitor the flow field in the rotating reference frame.

5 Conclusions

Experiments measuring the pressure fluctuation of compressor tip flow were performed to study the circumferential characteristics of RI phenomena. The main conclusions are listed below.

1. A calibration and prediction method of higher circumferential mode order is proposed, using two sets of evenly spaced angular sensors during measurements to overcome mode aliasing phenomena and capture flow characteristics of relatively high circumferential mode order using much fewer sensors. By using this method, theoretically with the numbers of sensors being$N_1 + N_2$, the maximum resolved mode order is significantly enhanced from $(N_1 + N_2)/2$ to $LCM(N_1, N_2)/2$(from 19 to 90 in this paper). The validation of this method was accomplished by analysing the mode order component at BPF. Further discussion shows that the calibration process performs better in mode components of single frequency (RF& BPF) than in wide-band hump regions (RIF) when some aliasing region still exists.
2. By using the calibration method together with the traditional mode decomposition method, the unsteady pressure fluctuation under different flow conditions was



(a) Source frequency **(b)** Source rotating speed

Fig. 15 Least square fitting results of the aerodynamic source model

analysed. A wide-band hump increases in the spectra beyond the 1st BPF as the flow rate decreases. Gradually, a characteristic RIF side-by-side peak forms in the PSD spectra and it keeps moving to a relatively low frequency. Calibrated mode analysis showed a structure of continuous numbered mode orders arranged in order at RIF, whereas a dominant mode order for each peak of the origin spectrum can be identified only in relatively small flow conditions.

3. Under small flow conditions, the wide-band hump characteristic changes, with an obvious single peak and fixed frequency range near 700 Hz. The spectrum structure involves only the mode components of 1, 2 and 3, which corresponds to the resonance between the RI flow and the natural frequency of the rotor. It is still uncertain why the resonance observed under low flow rate conditions with RI frequency is different from the natural frequency of the rotor.
4. A rotating aerodynamic source model is used to explain the characteristic spectrum of RIF. The RI phenomenon is assumed generated as an aerodynamic source with a certain source frequency, rotating speed and contains only a single circumferential mode order. The source frequency of RI gets lower with the flow rate decreases and become almost zero in small flow rate conditions. While the rotating speed changes slightly and stays around 40–45% rotor speed. The characteristic RIF side-by-side peak in PSD is considered as the effect of RI source being rotating. Further validation work of this model is needed, and results in numerical work are considered might be useful for it can monitor the flow field in the rotating reference frame.

Acknowledgements This work is supported by the National Science Foundation of China (grant no. 11202132).

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